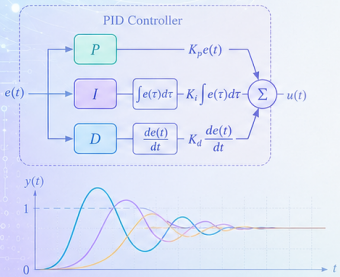
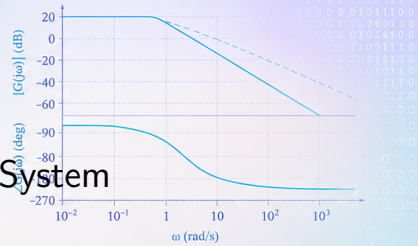
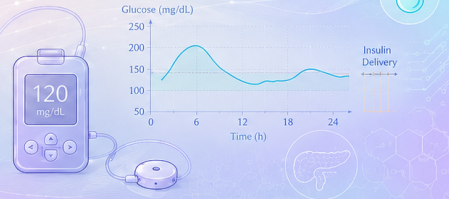
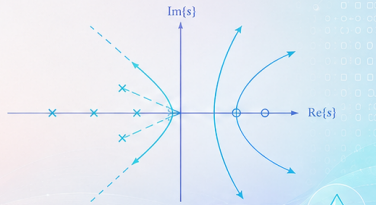


Transfer Function of System

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$$G(s) = \frac{Y(s)}{U(s)} \quad a_n \frac{d^n y(t)}{dt^n} + \dots + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_m \dots + b_m \frac{d^m u(t)}{dt^m} + \dots + b_1 \frac{du(t)}{dt} + b_0 u(t)$$



LTI System Response and Convolution Equation

- Consider the LTI state-space model:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t), \quad \mathbf{x}(t_0) = \mathbf{x}_0,$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t).$$

LTI System Response and Convolution Equation

- Consider the LTI state-space model:

$$\dot{\mathbf{x}}(t) = A\mathbf{x}(t) + B\mathbf{u}(t), \quad \mathbf{x}(t_0) = \mathbf{x}_0,$$

$$\mathbf{y}(t) = C\mathbf{x}(t) + D\mathbf{u}(t).$$

- The system output satisfies the **convolution equation**:

$$\mathbf{y}(t) = Ce^{A(t-t_0)}\mathbf{x}_0 + \int_{t_0}^t Ce^{A(t-\tau)}B\mathbf{u}(\tau) d\tau + D\mathbf{u}(t).$$

LTI System Response and Convolution Equation

- Consider the LTI state-space model:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t), \quad \mathbf{x}(t_0) = \mathbf{x}_0,$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t).$$

- The system output satisfies the **convolution equation**:

$$\mathbf{y}(t) = \mathbf{C}e^{\mathbf{A}(t-t_0)}\mathbf{x}_0 + \int_{t_0}^t \mathbf{C}e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau) d\tau + \mathbf{D}\mathbf{u}(t).$$

- The output $\mathbf{y}(t)$ satisfies linear properties:
 - Case 1: Zero initial state $\mathbf{x}(0) = 0$: the output $\mathbf{y}(t)$ is linear in input $\mathbf{u}(t)$
 - Case 2: Zero input $\mathbf{u}(t) = 0$: the output $\mathbf{y}(t)$ is linear in initial state $\mathbf{x}(0)$

Observations:

- ▶ Due to the linearity of matrix–vector multiplication and integration, the output is **jointly linear** in the initial condition $\mathbf{x}_0$ and the input $\mathbf{u}(t)$.
- ▶ The objective of control design is to choose the input $\mathbf{u}(t)$ to shape the output $\mathbf{y}(t)$, for example, to achieve regulation or reference tracking with limited overshoot, oscillations, and sensitivity to noise.
- ▶ Using the convolution equation directly for control-system analysis and design can be challenging.
- ▶ Therefore, we seek a simpler relationship between $\mathbf{u}(t)$ and $\mathbf{y}(t)$ by transforming the LTI model from the time domain to the frequency domain using the **Laplace transform**.

Complex Numbers $\mathbb{C}$

- ▶ The **space of real numbers** is denoted by $\mathbb{R}$.
- ▶ The **space of complex numbers** is denoted by $\mathbb{C}$.
- ▶ A **complex number** has the form

$$s = \sigma + j\omega,$$

where

$$\sigma, \omega \in \mathbb{R}, \quad j = \sqrt{-1}.$$

- ▶ In Cartesian coordinates,

$$s = \sigma + j\omega.$$

- ▶ The **real part** of s is

$$\operatorname{Re}(s) = \sigma.$$

- ▶ The **imaginary part** of s is

$$\operatorname{Im}(s) = \omega.$$

Complex Numbers: Polar Form

- In polar coordinates,

$$s = re^{j\theta} = r(\cos \theta + j \sin \theta).$$

- The **magnitude** of s is

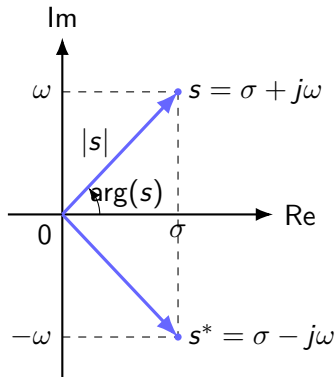
$$|s| = r = \sqrt{\sigma^2 + \omega^2}.$$

- The **phase** of s is

$$\arg(s) = \angle s = \theta = \text{atan2}(\omega, \sigma).$$

- The **complex conjugate** of

$$s = \sigma + j\omega$$



Complex Polynomial

- ▶ A **complex polynomial** of order n is a function

$$a : \mathbb{C} \rightarrow \mathbb{C}.$$

- ▶ It is written as

$$a(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_2 s^2 + a_1 s + a_0,$$

where

$$a_0, a_1, \dots, a_n \in \mathbb{C}$$

are constants.

- ▶ A **root** of the complex polynomial $a(s)$ is a number $\lambda \in \mathbb{C}$ such that

$$a(\lambda) = 0.$$

Complex Polynomial: Roots and Multiplicity

- ▶ A root λ of **multiplicity** m of a complex polynomial $a(s)$ satisfies

$$\lim_{s \rightarrow \lambda} \frac{a(s)}{(s - \lambda)^m} < \infty.$$

- ▶ **Fundamental theorem of algebra:**
A complex polynomial $a(s)$ of degree n has exactly n roots, counting multiplicities.
- ▶ Therefore, it can be factorized as

$$a(s) = a_n s^n + \cdots + a_0 = a_n (s - \lambda_1)(s - \lambda_2) \cdots (s - \lambda_n),$$

where

$$\lambda_1, \lambda_2, \dots, \lambda_n$$

are the n roots of $a(s)$.

Complex Polynomial with Real Coefficients

- ▶ A complex polynomial of order n with real coefficients is written as

$$a(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_2 s^2 + a_1 s + a_0,$$

where

$$a_0, a_1, \dots, a_n \in \mathbb{R}.$$

- ▶ The roots of a polynomial with real coefficients are either real,

$$\lambda = \sigma,$$

or occur in complex conjugate pairs,

$$\lambda = \sigma \pm j\omega.$$

Factorization of a Real-Coefficient Polynomial

- ▶ Every complex polynomial with real coefficients can be factorized into polynomials of degree one or two:

$$\begin{aligned} a(s) &= a_n s^n + \cdots + a_0 \\ &= a_n \prod_{i=1}^{n_1} (s - \lambda_i) \prod_{k=1}^{n_2} (s^2 + 2\zeta_k \omega_k s + \omega_k^2). \end{aligned}$$

- ▶ Here,
 - ▶ n_1 is the number of real roots;
 - ▶ n_2 is the number of complex conjugate root pairs.

Vieta's Formulas

- ▶ **Vieta's formulas** relate the polynomial coefficients a_i to the roots λ_i .
- ▶ The sum of all roots is

$$\sum_{i=1}^n \lambda_i = -\frac{a_{n-1}}{a_n}.$$

- ▶ The product of all roots is

$$\prod_{i=1}^n \lambda_i = (-1)^n \frac{a_0}{a_n}.$$

- ▶ More generally,

$$\sum_{1 \leq i_1 < \dots < i_k \leq n} \prod_{j=1}^k \lambda_{i_j} = (-1)^k \frac{a_{n-k}}{a_n}.$$

Rational Function

- ▶ A rational function

$$F : \mathbb{C} \rightarrow \mathbb{C}$$

is a ratio of two complex polynomials:

$$F(s) = \frac{b(s)}{a(s)}.$$

- ▶ In expanded form,

$$F(s) = \frac{b_m s^m + \cdots + b_1 s + b_0}{a_n s^n + \cdots + a_1 s + a_0}.$$

- ▶ Rational functions remain rational functions under
 - ▶ addition;
 - ▶ subtraction;
 - ▶ multiplication;
 - ▶ division, except division by zero.

Characteristic Equation, Zeros and Poles

- ▶ The **characteristic equation** of

$$F(s) = \frac{b(s)}{a(s)}$$

is

$$a(s) = 0.$$

- ▶ A **zero** $z \in \mathbb{C}$ of $F(s)$ is a root of the numerator:

$$b(z) = 0.$$

- ▶ A **pole** $p \in \mathbb{C}$ of $F(s)$ is a root of the characteristic equation:

$$a(p) = 0.$$

Pole-Zero Form

- ▶ The **pole-zero form** of a rational function $F(s)$ is

$$F(s) = \frac{b_ms^m + \cdots + b_1s + b_0}{a_ns^n + \cdots + a_1s + a_0}.$$

- ▶ By factorizing the numerator and denominator,

$$F(s) = k \frac{(s - z_1)(s - z_2) \cdots (s - z_m)}{(s - p_1)(s - p_2) \cdots (s - p_n)},$$

where

$$k = \frac{b_m}{a_n}.$$

- ▶ Here,
 - ▶ $z_1, \dots, z_m$ are the zeros of $F(s)$;
 - ▶ $p_1, \dots, p_n$ are the poles of $F(s)$.

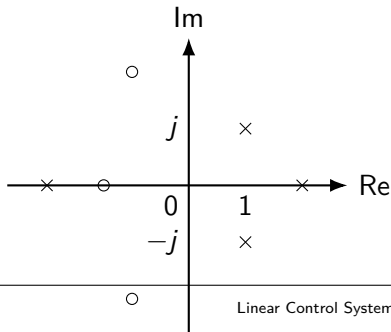
Pole-Zero Map

- ▶ A **pole-zero map** is a plot of the poles and zeros of $F(s)$ in the complex s -plane.
- ▶ Consider

$$F(s) = k \frac{(s + 1.5)(s + 1 + 2j)(s + 1 - 2j)}{(s + 2.5)(s - 2)(s - 1 - j)(s - 1 + j)}.$$

- ▶ The symbols used in a pole-zero map are

$\times$ = pole, $\circ$ = zero.



Example: Zeros and Poles

- Consider

$$F(s) = \frac{2s + 1}{3s^2 + 2s + 1}.$$

- The numerator is zero when

$$2s + 1 = 0.$$

- Therefore, $F(s)$ has one zero:

$$z = -\frac{1}{2}.$$

- The roots of a quadratic polynomial

$$a(s) = a_2s^2 + a_1s + a_0$$

are

$$s = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_2a_0}}{2a_2}.$$

Example: Poles and Pole-Zero Form

- For the denominator

$$3s^2 + 2s + 1 = 0,$$

the two poles are

$$p_1 = -\frac{1}{3} + j\frac{\sqrt{2}}{3},$$

and

$$p_2 = -\frac{1}{3} - j\frac{\sqrt{2}}{3}.$$

- Therefore, the pole-zero form of $F(s)$ is

$$F(s) = \frac{2(s - z)}{3(s - p_1)(s - p_2)}.$$

- Here,

$$z = -\frac{1}{2}, \quad p_{1,2} = -\frac{1}{3} \pm j\frac{\sqrt{2}}{3}.$$

Laplace Transform and Inverse Laplace Transform

- The **Laplace transform** $F(s)$ of a function $f(t)$ is

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt,$$

where

$$s = \sigma + j\omega$$

is a complex number.

- The **inverse Laplace transform** $f(t)$ of a function $F(s)$ is

$$f(t) = \mathcal{L}^{-1}\{F(s)\} = \frac{1}{2\pi j} \lim_{\omega \rightarrow \infty} \int_{\sigma - j\omega}^{\sigma + j\omega} F(s)e^{st} ds,$$

where σ is greater than the real part of all singularities of $F(s)$.

Inverse Laplace Transform Using Residues

► **Cauchy's Residue Theorem:**

If $F(s)$ is a strictly proper rational function, then

$$f(t) = \mathcal{L}^{-1}\{F(s)\} = \sum_{\substack{s \text{ is a} \\ \text{pole of } F(s)}} \text{Res} [F(s)e^{st}] .$$

► Equivalently,

$$f(t) = \sum_{\substack{s \text{ is a} \\ \text{pole of } F(s)}} (\text{residue of } F(s)e^{st} \text{ at } s) .$$

Laplace Transform Properties: Linearity

- The Laplace transform is **linear**:

$$\begin{aligned}\mathcal{L}\{\alpha f(t) + \beta g(t)\} &= \int_0^{\infty} (\alpha f(t) + \beta g(t)) e^{-st} dt \\ &= \alpha \int_0^{\infty} f(t) e^{-st} dt + \beta \int_0^{\infty} g(t) e^{-st} dt \\ &= \alpha \mathcal{L}\{f(t)\} + \beta \mathcal{L}\{g(t)\}.\end{aligned}$$

Linearity property

$$\mathcal{L}\{\alpha f(t) + \beta g(t)\} = \alpha F(s) + \beta G(s).$$

Laplace Transform Properties: Convolution

- For $f(t)$ and $g(t)$ supported on $t \in [0, \infty)$, their **convolution** is

$$(f * g)(t) = \int_0^t f(\tau)g(t - \tau) d\tau.$$

- Convolution in the time domain becomes multiplication in the complex domain:

$$\mathcal{L}\{(f * g)(t)\} = \mathcal{L}\{f(t)\}\mathcal{L}\{g(t)\}.$$

- Therefore,

$$\boxed{\mathcal{L}\{(f * g)(t)\} = F(s)G(s)}.$$

Convolution Property: Derivation

$$\mathcal{L}\{(f * g)(t)\} = \int_0^\infty \int_0^t f(\tau)g(t - \tau)e^{-st} d\tau dt.$$

Let

$$\mu = t - \tau, \quad t = \mu + \tau.$$

Then

$$\mathcal{L}\{(f * g)(t)\} = \int_0^\infty \int_{-\tau}^\infty f(\tau)g(\mu)e^{-s\tau}e^{-s\mu} d\mu d\tau.$$

Since

$$g(\mu) = 0, \quad \mu < 0,$$

we obtain

$$\begin{aligned}\mathcal{L}\{(f * g)(t)\} &= \left(\int_0^\infty f(\tau)e^{-s\tau} d\tau \right) \left(\int_0^\infty g(\mu)e^{-s\mu} d\mu \right) \\ &= \mathcal{L}\{f(t)\}\mathcal{L}\{g(t)\}.\end{aligned}$$

Laplace Transform Properties: Differentiation

- ▶ The Laplace transform of the first derivative is

$$\mathcal{L}\left\{\frac{d}{dt}x(t)\right\} = s\mathcal{L}\{x(t)\} - x(0).$$

- ▶ If

$$X(s) = \mathcal{L}\{x(t)\},$$

then

$$\boxed{\mathcal{L}\{\dot{x}(t)\} = sX(s) - x(0)}.$$

- ▶ More generally,

$$\mathcal{L}\{x^{(n)}(t)\} = s^n X(s) - \sum_{k=1}^n s^{n-k} x^{(k-1)}(0).$$

Differentiation Property: Proof

- Consider

$$\int_0^{\infty} \frac{d}{dt} (x(t)e^{-st}) dt.$$

- Evaluating the integral,

$$\int_0^{\infty} \frac{d}{dt} (x(t)e^{-st}) dt = x(t)e^{-st} \Big|_0^{\infty} = -x(0).$$

- Using the product rule,

$$\begin{aligned} & \int_0^{\infty} \frac{d}{dt} (x(t)e^{-st}) dt \\ &= \int_0^{\infty} \dot{x}(t)e^{-st} dt + \int_0^{\infty} x(t) \frac{d}{dt} (e^{-st}) dt \\ &= \mathcal{L}\{\dot{x}(t)\} - s\mathcal{L}\{x(t)\}. \end{aligned}$$

Laplace Transform Properties: Integration

- ▶ The Laplace transform of a time-domain integral is

$$\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{1}{s}\mathcal{L}\{f(t)\}.$$

- ▶ If

$$F(s) = \mathcal{L}\{f(t)\},$$

then

$$\boxed{\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{F(s)}{s}}.$$

- ▶ Note that

$$\frac{d}{dt}\left(\int_0^t f(\tau) d\tau\right) = f(t).$$

Laplace Transform of an Exponential Function

- Consider the function

$$f(t) = e^{at}.$$

- Its Laplace transform is

$$\begin{aligned}\mathcal{L}\{e^{at}\} &= \int_0^{\infty} e^{at} e^{-st} dt = \int_0^{\infty} e^{-(s-a)t} dt \\ &= -\frac{1}{s-a} e^{-(s-a)t} \bigg|_{t=0}^{t=\infty}.\end{aligned}$$

- Provided that

$$\operatorname{Re}(s) > a,$$

we obtain

$$\boxed{\mathcal{L}\{e^{at}\} = \frac{1}{s-a}}.$$

Delta Function

- ▶ The approximate impulse function is defined as

$$\delta_{\epsilon}(t) = \begin{cases} 0, & t < 0, \\ \frac{1}{\epsilon}, & 0 \leq t < \epsilon, \\ 0, & t \geq \epsilon. \end{cases}$$

- ▶ The ideal impulse is obtained in the limit:

$$\delta(t) = \lim_{\epsilon \rightarrow 0} \delta_{\epsilon}(t) = \begin{cases} \infty, & t = 0, \\ 0, & t \neq 0. \end{cases}$$

- ▶ For any function $f(t)$ continuous at $\tau \in (a, b)$, the shifting property is

$$\int_a^b f(t) \delta(t - \tau) dt = f(\tau).$$

Laplace Transform of the Delta Function

- The Laplace transform of the impulse function is

$$\begin{aligned}\mathcal{L}\{\delta(t)\} &= \int_0^{\infty} \delta(t) e^{-st} dt \\ &= e^{-st} \Big|_{t=0} \\ &= 1.\end{aligned}$$

- Therefore,

$$\boxed{\mathcal{L}\{\delta(t)\} = 1}.$$

Heaviside Step Function

- The Heaviside step function is

$$H(t) = \int_{-\infty}^t \delta(\tau) d\tau = \begin{cases} 1, & t \geq 0, \\ 0, & t < 0. \end{cases}$$

- Its Laplace transform is

$$\boxed{\mathcal{L}\{H(t)\} = \frac{1}{s}}.$$

Ramp and Parabola Functions

- The **ramp function** is

$$tH(t) = \begin{cases} t, & t \geq 0, \\ 0, & t < 0. \end{cases}$$

Its Laplace transform is

$$\mathcal{L}\{tH(t)\} = \frac{1}{s^2}.$$

- The **parabola function** is

$$\frac{t^2}{2}H(t) = \begin{cases} \frac{t^2}{2}, & t \geq 0, \\ 0, & t < 0. \end{cases}$$

Its Laplace transform is

$$\mathcal{L}\left\{\frac{t^2}{2}H(t)\right\} = \frac{1}{s^3}.$$

Laplace Transform Properties

Property	t -domain	s -domain
Linearity	$af(t) + bg(t)$	$aF(s) + bG(s)$
Convolution	$(f * g)(t)$	$F(s)G(s)$
Scaling, $a > 0$	$f(at)$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
s -domain derivative	$t^n f(t)$	$(-1)^n F^{(n)}(s)$
s -domain shift	$e^{at} f(t)$	$F(s - a)$

Laplace Transform Properties (Continued)

Property	t -domain	s -domain
Time-domain derivative	$f^{(n)}(t)$	$s^n F(s) - \sum_{k=1}^n s^{n-k} f^{(k-1)}(0)$
s -domain integration	$\frac{1}{t} f(t)$	$\int_s^\infty F(\sigma) d\sigma$
Time-domain integration	$\int_0^t f(\tau) d\tau$	$\frac{1}{s} F(s)$
Time-domain shift, $a > 0$	$f(t - a)H(t - a)$	$e^{-as} F(s)$

Multiplication Property

- Multiplication in the time domain corresponds to convolution in the complex domain:

$$f(t)g(t) \xleftrightarrow{\mathcal{L}} \frac{1}{2\pi j} \int_{\operatorname{Re}(\sigma)-j\infty}^{\operatorname{Re}(\sigma)+j\infty} F(\sigma)G(s-\sigma) d\sigma.$$

- Therefore,

$$\boxed{\mathcal{L}\{f(t)g(t)\} = \frac{1}{2\pi j} \int_{\operatorname{Re}(\sigma)-j\infty}^{\operatorname{Re}(\sigma)+j\infty} F(\sigma)G(s-\sigma) d\sigma.}$$

Common Laplace Transform Pairs

$f(t)$

$$\int_{-\infty}^t f(\tau) d\tau$$

Impulse function $\delta(t)$

$$e^{-at} \sin(\omega t)$$

$$e^{-at} \cos(\omega t)$$

$F(s)$

$$\frac{F(s)}{s} + \frac{1}{s} \int_{-\infty}^0 f(t) dt$$

$$1$$

$$\frac{\omega}{(s+a)^2 + \omega^2}$$
$$\frac{s+a}{(s+a)^2 + \omega^2}$$

Common Laplace Transform Pairs (Continued)

$$\frac{1}{\omega} \sqrt{(\alpha - a)^2 + \omega^2} e^{-at} \sin(\omega t + \phi) \quad \xleftrightarrow{\mathcal{L}} \quad \frac{s + \alpha}{(s + a)^2 + \omega^2},$$

where

$$\phi = \tan^{-1} \left(\frac{\omega}{\alpha - a} \right).$$

$$\frac{\omega_n}{\sqrt{1 - \zeta^2}} e^{-\zeta \omega_n t} \sin \left(\omega_n \sqrt{1 - \zeta^2} t \right) \quad \xleftrightarrow{\mathcal{L}} \quad \frac{\omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2},$$

where

$$\zeta < 1.$$

Common Laplace Transform Pairs: Step Responses

$$\frac{1}{a^2 + \omega^2} + \frac{1}{\omega\sqrt{a^2 + \omega^2}} e^{-at} \sin(\omega t - \phi) \xleftrightarrow{\mathcal{L}} \frac{1}{s[(s+a)^2 + \omega^2]},$$

where

$$\phi = \tan^{-1} \left(\frac{\omega}{-a} \right).$$

$$1 - \frac{1}{\sqrt{1 - \zeta^2}} e^{-\zeta\omega_n t} \sin \left(\omega_n \sqrt{1 - \zeta^2} t + \phi \right) \xleftrightarrow{\mathcal{L}} \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)},$$

where

$$\phi = \cos^{-1} \zeta, \quad \zeta < 1.$$

Common Laplace Transform Pairs: General Form

$$\frac{\alpha}{\alpha^2 + \omega^2} + \frac{1}{\omega} \sqrt{\frac{(\alpha - a)^2 + \omega^2}{\alpha^2 + \omega^2}} e^{-at} \sin(\omega t + \phi) \xleftrightarrow{\mathcal{L}} \frac{s + \alpha}{s[(s + a)^2 + \omega^2]}.$$

where

$$\phi = \tan^{-1} \left(\frac{\omega}{\alpha - a} \right) - \tan^{-1} \left(\frac{\omega}{-a} \right).$$

Initial and Final Value Theorems

Initial Value Theorem

Suppose that $f(t)$ has a Laplace transform $F(s)$. Then

$$\lim_{t \rightarrow 0} f(t) = \lim_{s \rightarrow \infty} sF(s).$$

Final Value Theorem

Suppose that $f(t)$ has a Laplace transform $F(s)$, and every pole of $F(s)$ is either in the open left-half plane or at the origin of $\mathbb{C}$. Then

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s).$$

Laplace Transform of an LTI ODE

- Consider an n th-order LTI ODE with zero initial conditions:

$$a_n \frac{d^n y(t)}{dt^n} + a_{n-1} \frac{d^{n-1} y(t)}{dt^{n-1}} + \cdots + a_0 y(t) = b_m \frac{d^m u(t)}{dt^m} + b_{m-1} \frac{d^{m-1} u(t)}{dt^{m-1}} + \cdots + b_0 u(t)$$

- Let

$$Y(s) = \mathcal{L}\{y(t)\}, \quad U(s) = \mathcal{L}\{u(t)\}.$$

- Recall that

$$\mathcal{L}\left\{\frac{d^n y(t)}{dt^n}\right\} = s^n Y(s) - \sum_{k=1}^n s^{n-k} \left. \frac{d^{k-1} y(t)}{dt^{k-1}} \right|_{t=0}.$$

- Under zero initial conditions,

$$\mathcal{L}\left\{\frac{d^n y(t)}{dt^n}\right\} = s^n Y(s).$$

Laplace Transform of an LTI ODE

- ▶ Applying the Laplace transform to the differential equation gives

$$(a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0) Y(s) = (b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0) U(s).$$

- ▶ Rearranging the equation,

$$\frac{Y(s)}{U(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0}.$$

- ▶ The **transfer function** is therefore

Transfer Function

$$G(s) = \frac{Y(s)}{U(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0}.$$

Transfer Function

Transfer Function

The **transfer function** $G(s)$ of a single-input single-output LTI system is the ratio of the Laplace transform $Y(s)$ of the output $y(t)$ to the Laplace transform $U(s)$ of the input $u(t)$, with zero initial conditions:

$$G(s) = \frac{Y(s)}{U(s)}.$$

Relative Degree

For the transfer function

$$G(s) = \frac{b_ms^m + b_{m-1}s^{m-1} + \dots + b_0}{a_ns^n + a_{n-1}s^{n-1} + \dots + a_0},$$

the **relative degree** is $r = n - m$.

Assuming that there are no pole-zero cancellations, it represents the difference between the number of poles and the number of zeros.

Classification Based on Relative Degree

► If

$$r > 0,$$

the transfer function is called **strictly proper**.

► If

$$r = 0,$$

the transfer function is called **proper**.

► If

$$r < 0,$$

the transfer function is called **improper**.

► An improper transfer function cannot be represented by a standard finite dimensional causal state-space model without introducing additional dynamics.

Example

- ▶ A vehicle with position $p(t)$ and acceleration input $u(t)$ satisfies

$$m\ddot{p}(t) = u(t).$$

- ▶ Taking the Laplace transform under zero initial conditions,

$$ms^2P(s) = U(s).$$

- ▶ The transfer function of the system is

$$G(s) = \frac{P(s)}{U(s)} = \frac{1}{ms^2}.$$

- ▶ The transfer function is **strictly proper**, with relative degree

$$r = 2.$$

Example: Second-Order LTI ODE

- ▶ Consider a second-order system with natural frequency ω_n , damping ratio ζ , and input $u(t)$:

$$\ddot{y}(t) + 2\zeta\omega_n\dot{y}(t) + \omega_n^2y(t) = u(t).$$

- ▶ Applying the Laplace transform gives

$$s^2Y(s) - sy(0) - \dot{y}(0) + 2\zeta\omega_n(sY(s) - y(0)) + \omega_n^2Y(s) = U(s).$$

- ▶ Under zero initial conditions,

$$y(0) = 0, \quad \dot{y}(0) = 0,$$

the transfer function is

$$G(s) = \frac{Y(s)}{U(s)} = \frac{1}{s^2 + 2\zeta\omega_ns + \omega_n^2}.$$

Second-Order LTI ODE: Total Response

The total response in the Laplace domain is

$$Y(s) = \underbrace{\frac{(s + 2\zeta\omega_n)y(0) + \dot{y}(0)}{s^2 + 2\zeta\omega_ns + \omega_n^2}}_{\text{natural response}} + \underbrace{G(s)U(s)}_{\text{forced response}}.$$

- ▶ The **natural response** depends on the initial conditions.
- ▶ The **forced response** is produced by the external input $u(t)$.
- ▶ Under zero initial conditions, only the forced response remains:

$$Y(s) = G(s)U(s).$$

Transfer Function of State-Space Model

- ▶ Consider an LTI system in state-space form:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t),$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t).$$

- ▶ Applying the Laplace transform gives

$$s\mathbf{X}(s) - \mathbf{x}(0) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}\mathbf{U}(s),$$

$$\mathbf{Y}(s) = \mathbf{C}\mathbf{X}(s) + \mathbf{D}\mathbf{U}(s).$$

- ▶ Rearranging the state equation,

$$(s\mathbf{I} - \mathbf{A})\mathbf{X}(s) = \mathbf{x}(0) + \mathbf{B}\mathbf{U}(s).$$

- ▶ Therefore,

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}(0) + (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{U}(s).$$

Transfer Function of State-Space Model

- ▶ Substituting $\mathbf{X}(s)$ into the output equation gives

$$\mathbf{Y}(s) = \underbrace{C(sI - A)^{-1}\mathbf{x}(0)}_{\text{natural response}} + \underbrace{[C(sI - A)^{-1}B + D]\mathbf{U}(s)}_{\text{forced response}}.$$

- ▶ The **natural response** is produced by the initial condition $\mathbf{x}(0)$.
- ▶ The **forced response** is produced by the external input $\mathbf{u}(t)$.
- ▶ Under zero initial conditions,

$$\mathbf{x}(0) = 0,$$

and hence

$$\mathbf{Y}(s) = G(s)\mathbf{U}(s).$$

Transfer Function

$$G(s) = C(sI - A)^{-1}B + D$$

Example

- Consider a SISO LTI system with the state-space matrices

$$A = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

$$C = [1 \quad 0], \quad D = 0.$$

- The transfer function is

$$G(s) = C(sI - A)^{-1}B + D.$$

- First,

$$sI - A = \begin{bmatrix} s & -1 \\ 1 & s + 2 \end{bmatrix}.$$

Example: Transfer-Function Calculation

$$\begin{aligned} G(s) &= [1 \quad 0] \begin{bmatrix} s & -1 \\ 1 & s+2 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 0 \\ &= [1 \quad 0] \frac{1}{s^2 + 2s + 1} \begin{bmatrix} s+2 & 1 \\ -1 & s \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ &= \frac{1}{s^2 + 2s + 1} [1 \quad 0] \begin{bmatrix} 1 \\ s \end{bmatrix} \\ &= \boxed{\frac{1}{s^2 + 2s + 1}}. \end{aligned}$$

Since $s^2 + 2s + 1 = (s + 1)^2$, $G(s)$ can also be written as $G(s) = \frac{1}{(s+1)^2}$.

Controllable Canonical Form

- Consider general n th-order transfer function, where some coefficients b_i may be zero:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{b_n s^n + b_{n-1} s^{n-1} + \dots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0}.$$

- Introduce an auxiliary variable $Z(s)$ and write

- Choose
$$G(s) = \frac{Y(s)/Z(s)}{U(s)/Z(s)}.$$

$$\frac{U(s)}{Z(s)} = s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0,$$

and

$$\frac{Y(s)}{Z(s)} = b_n s^n + b_{n-1} s^{n-1} + \dots + b_1 s + b_0.$$

Controllable Canonical Form: Time-Domain Equations

Taking the inverse Laplace transform gives

$$y(t) = b_n z^{(n)}(t) + b_{n-1} z^{(n-1)}(t) + \cdots + b_1 \dot{z}(t) + b_0 z(t),$$

$$u(t) = z^{(n)}(t) + a_{n-1} z^{(n-1)}(t) + \cdots + a_1 \dot{z}(t) + a_0 z(t).$$

► From the second equation,

$$z^{(n)}(t) = -a_0 z(t) - a_1 \dot{z}(t) - \cdots - a_{n-1} z^{(n-1)}(t) + u(t).$$

► Select the state variables as

$$x_1 = z, \quad x_2 = \dot{z}, \quad \dots, \quad x_n = z^{(n-1)}.$$

Controllable Canonical Form: State Equation

The state-variable relationships are

$$\dot{x}_1 = x_2,$$

$$\dot{x}_2 = x_3,$$

$$\vdots$$

$$\dot{x}_{n-1} = x_n,$$

$$\dot{x}_n = -a_0x_1 - a_1x_2 - \cdots - a_{n-1}x_n + u.$$

Therefore,

$$\dot{\mathbf{x}} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ -a_0 & -a_1 & \cdots & -a_{n-2} & -a_{n-1} \end{bmatrix}}_A \mathbf{x} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}}_B u.$$

Controllable Canonical Form: Output Equation

Since

$$z^{(n)} = -a_0x_1 - a_1x_2 - \cdots - a_{n-1}x_n + u,$$

the output equation becomes

$$\begin{aligned} y &= b_n z^{(n)} + b_{n-1}x_n + \cdots + b_1x_2 + b_0x_1 \\ &= (b_0 - a_0b_n)x_1 + (b_1 - a_1b_n)x_2 + \cdots + (b_{n-1} - a_{n-1}b_n)x_n + b_nu. \end{aligned}$$

Thus,

$$y = \underbrace{\begin{bmatrix} b_0 - a_0b_n & b_1 - a_1b_n & \cdots & b_{n-1} - a_{n-1}b_n \end{bmatrix}}_C \mathbf{x} + \underbrace{b_n}_D u.$$

Controllable canonical realization

Thank You

Questions and Discussion